

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-1122

Fair Oaks 138 kV
October 04, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1122. The Interconnection Customer proposes a 228.7 MW wind facility interconnecting at the Fair Oaks 138 kV substation in the Pennsylvania area. The request was received on December 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 331 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,926,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Collierville / Sweet 138 kV L	Fair Oaks 345 kV/138 kV Transformer	129.5	537	100.0
Sweet / Nolensville 138 kV	Far Oaks 345 kV/138 kV XFMR	129.2	542	100.0
Nolensville / Coon Rapids li 138 kV	Fair Oaks 345/138 kV Autotransformer	128.8	546	100.0
High Street / Coon Rapids li 138 kV	Fair Oaks 345 kV/138 kV XFMR	128.6	550	100.0
Lynbrook / High St 138 kV	Fair Oaks 345/138kV Autotransformer	128.2	554	100.0
Lynbrook / Fair Oaks 138 kV	Fair Oaks 345/138 kV Autotransformer	127.9	558	100.0
Fair Oaks 345/138kV TX	Fair aks / Lynbrook 138 kV	114.2	625	100.0
W Waynesboro 345/138 kV TX	W Waynesboro / Pineville 345 kV	111.7	1,046	11.3
Mtn View / Antrim 138 kV	Fair Oaks / Chambersburg Substation 345 kV	111.6	434	9.8
Gravel Ridge / Shepherdstown 138 kV	Urban Substation / American 345 kV line	111.3	605	5.5
Chambersburg / North Lewiston 138 kV line	Chambersburg / West Waynesboro 345 kV L	111.2	692	8.0
Gravel Ridge / Dam No. 4 Hydro Sta 138 kV	Urban Substation / American 345 kV	111.2	612	5.5
Lourich / Mountain View 138 kV line	Fair Oaks / Chambersbug 345 kV L	110.9	447	9.8
Chambersburg 345 kV/138 kV Transformer Bank	W Waynesboro / Chambersburg 345 kV line	106.2	864	10.0
M105 / Otsego 138 kV L	Chambersburg / West Waynesboro 345 kV	106.0	694	5.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Carrel 345-138 kV Auto	Chambersburg / Fair Oaks 345 kV L	104.9	1,323	29.0
M105 / Wyoming 138 kV	W Waynesboro / Chambersburg 345 kV L	103.6	708	5.8
Wyoming Substation / N Lewiston 138 kV	W Waynesboro / Chambersburg 345 kV	103.4	726	5.8
Carrel / Lourich 138 kV	Fir Oaks / Chambersburg 345 kV	102.5	482	9.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Collierville / Sweet 138 kV	Rebuild line with higher-capacity conductor	\$3,296,000	31% (\$1,024,000)
Nolensville / Sweet Substation 138 kV	Rebuild line with higher-capacity conductor	\$10,342,000	31% (\$3,213,000)
Nolensville / Coon Rapids II 138 kV	Rebuild line with higher-capacity conductor	\$26,696,000	31% (\$8,294,000)
High Street / Coon Rapids II 138 kV	Rebuild line with higher-capacity conductor	\$9,017,000	31% (\$2,802,000)
Lynbrook / High St 138 kV	Rebuild line with higher-capacity conductor	\$3,588,000	31% (\$1,115,000)
Fair Oaks / Lynbrook 138 kV	Rebuild line with higher-capacity conductor	\$6,682,000	31% (\$2,076,000)
Fair Oaks 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,581,000	31% (\$4,841,000)
W Waynesboro 345/138 kV TX	Replace transformer with higher-rated unit	\$15,061,000	5% (\$695,000)
Mtn View / Antrim 138 kV	Rebuild line with higher-capacity conductor	\$8,905,000	6% (\$548,000)
Shepherdstown Substation / Gravel Ridge 138 kV	Rebuild line with higher-capacity conductor	\$3,095,000	4% (\$141,000)
Chambersburg / N Lewiston 138 kV	Rebuild line with higher-capacity conductor	\$5,018,000	6% (\$282,000)
Gravel Ridge / Dam No. 4 Hydro Sta 138 kV	Rebuild line with higher-capacity conductor	\$7,288,000	4% (\$331,000)
Lourich / Mountain View 138 kV line	Rebuild line with higher-capacity conductor	\$2,722,000	6% (\$168,000)
Chambersburg 345/138kV XFMR	Replace transformer with higher-rated unit	\$13,856,000	5% (\$706,000)
Otsego / M105 Substation 138 kV	Reconductor line	\$1,502,000	4% (\$62,000)
Carrel 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$19,047,000	8% (\$1,446,000)
M105 / Wyoming 138 kV	Reconductor line	\$434,000	4% (\$18,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wyoming Substation / N Lewiston 138 kV	Reconductor line	\$336,000	4% (\$14,000)
Carrel / Louich 138 kV line	Reconductor line	\$2,429,000	6% (\$150,000)
	Total allocated to project		\$27,926,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.